

A PROJECT REPORT

on

Unit Trade

As a part of

Mobile Application & Development

[CE0630]

Submitted by

Rishabh Chopda (IU2341230409)

Daxa Antiya (IU2341230468)

Viren Shah (IU2341230433)

In fulfillment for the award of the degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE ENGINEERING



INSTITUTE OF TECHNOLOGY AND ENGINEERING

INDUS UNIVERSITY CAMPUS, RANCHARDA, VIA-THALTEJ

AHMEDABAD-382115, GUJARAT, INDIA,

WEB: www.indusuni.ac.in

1. Introduction

UniTrade is a mobile-based marketplace application developed using Android Studio. The application allows users to buy, sell, and communicate with other users through an integrated messaging system. It is designed to provide a simple and efficient platform for peer-to-peer trading.

2. Objectives

- To develop a user-friendly marketplace application
- To enable users to buy and sell products easily
- To implement a real-time-like chat system using local database
- To provide category-based filtering of products
- To enhance user experience with modern UI design

3. Technologies Used

3.1 Frontend

- XML Layouts
- Material UI Components

3.2 Backend

- Java (Android)

3.3 Database

- SQLite

3.4 Tools

- Android Studio
- Emulator / Physical Device

4. System Architecture

The application follows a simple architecture:

User Interface (XML) → Activity (Java) → DBHelper → SQLite Database

- XML handles design
- Java handles logic
- DBHelper manages database operations

5. Features of the Application

- User Registration and Login
- Add (Sell) Products
- View (Buy) Products
- Category Filtering
- Product Details View
- Contact Seller (Chat System)
- Profile Management
- Settings (Language & Theme)
- Additional Modules (Lost & Found, Events – Coming Soon)

6. Database Design

6.1 Users Table

Field	Description
id	Primary Key
name	User Name
email	Unique Email
password	User Password
image	Profile Image

6.2 Products Table

Field	Description
id	Product ID

email	Seller Email
name	Product Name
price	Product Price
description	Product Description
category	Product Category
image	Product Image

6.3 Messages Table

Field	Description
id	Message ID
sender	Sender Email
receiver	Receiver Email
message	Message Content
time	Timestamp

7. Module Description

7.1 Login & Register Module

Handles user authentication using SQLite database.

7.2 Home Module

Displays main dashboard with options:

- Buy
- Sell
- Lost & Found
- Events

7.3 Buy Module

- Displays all products
- Uses ListView + Adapter
- Supports category filtering

7.4 Sell Module

- Allows users to upload product
- Includes image selection using storage

7.5 Product Details Module

- Shows full product info
- Includes “Contact Seller” option

7.6 Chat Module

- Enables messaging between users
- Messages stored in SQLite
- Works like basic WhatsApp chat

7.7 Profile Module

- Displays user info
- Allows profile update

7.8 Settings Module

- Language selection (Localization)
- Theme toggle (Light/Dark Mode)

8. UI/UX Design

- Material Design principles used
- Card-based layout for products
- Consistent color theme
- Responsive layouts
- Clean and modern interface

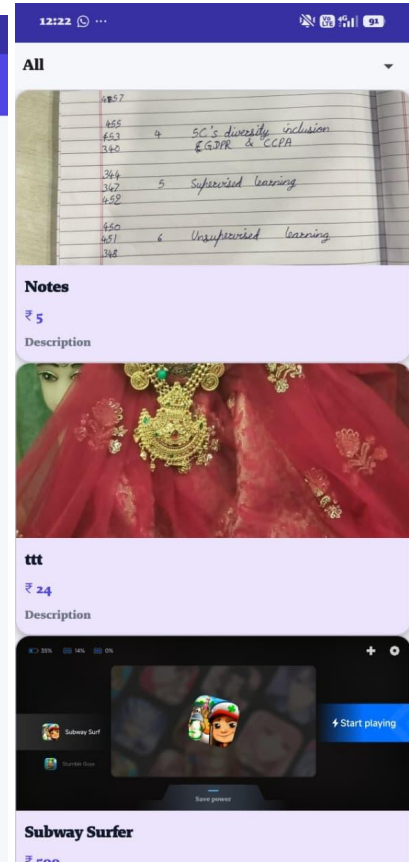
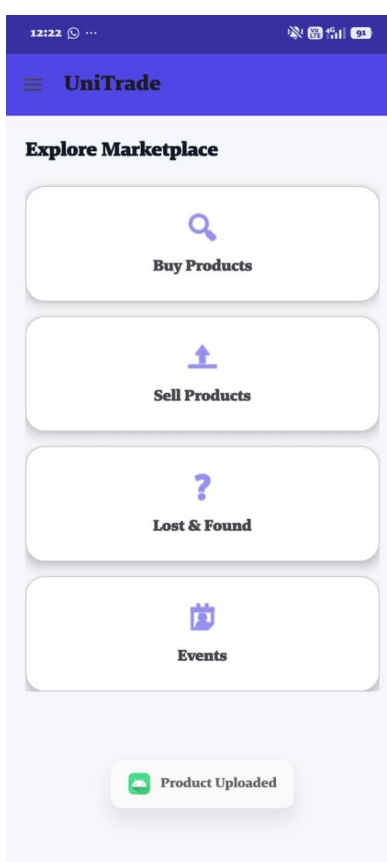
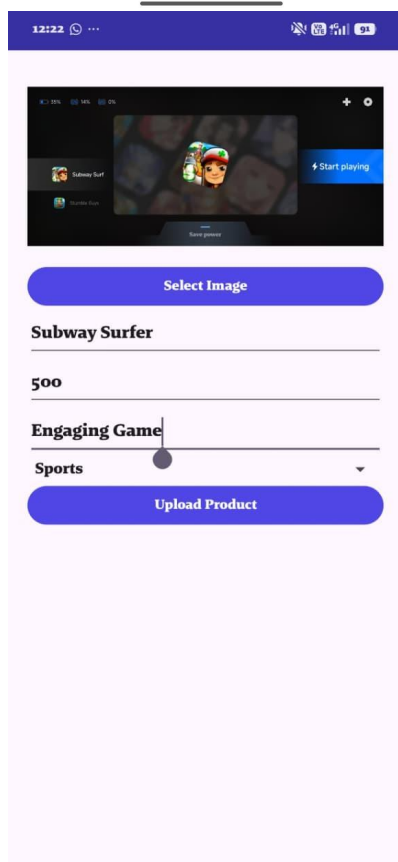
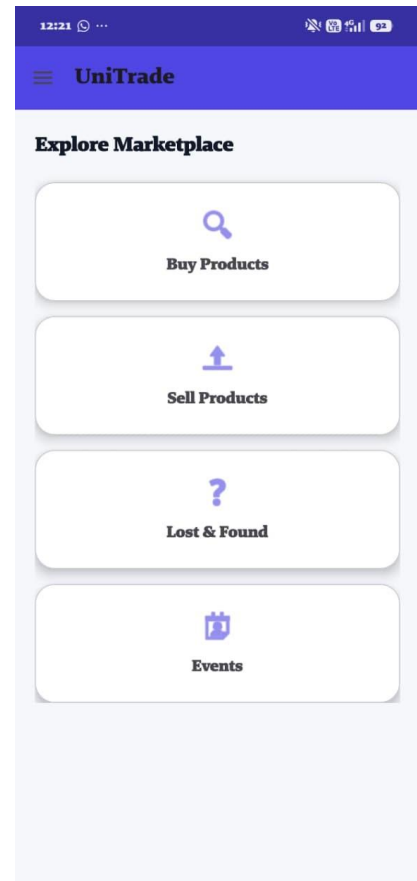
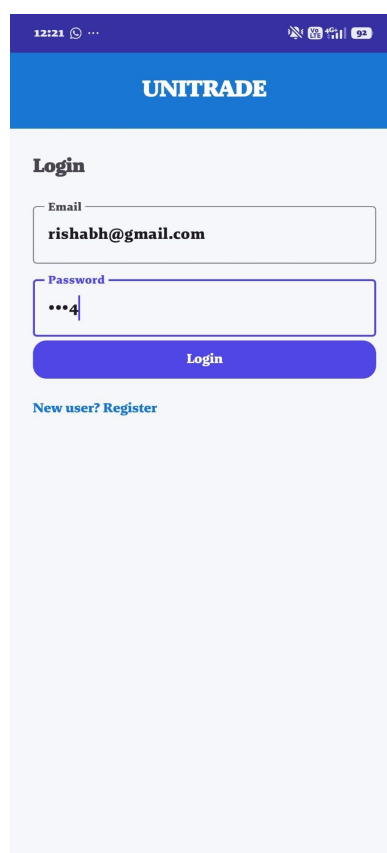
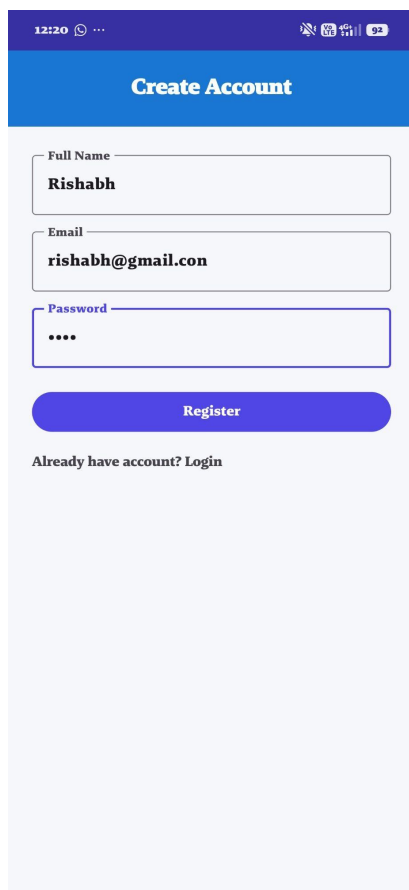
9. Key Concepts Used

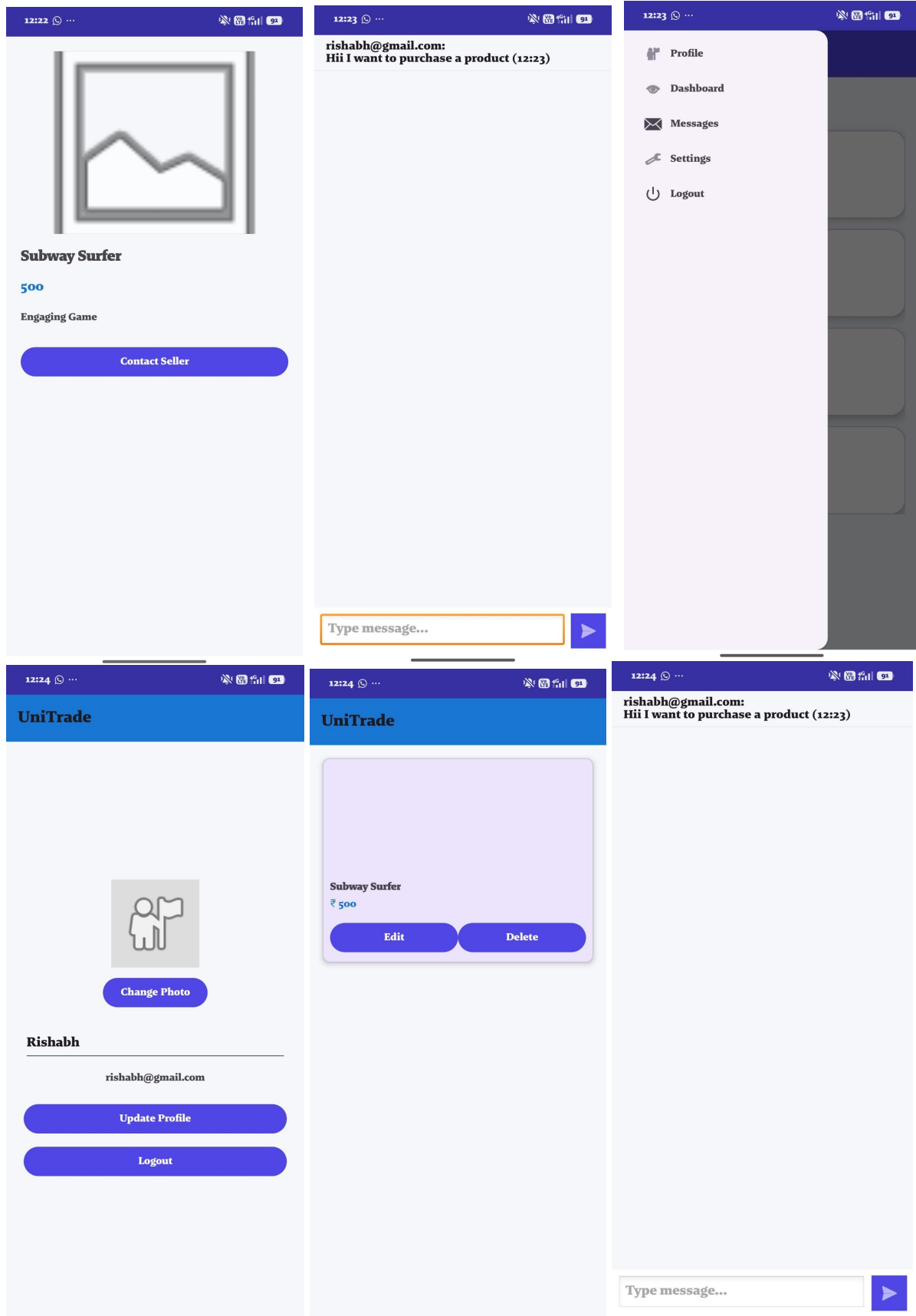
- **SQLite Database** – For offline data storage
- **Intents** – For activity navigation
- **Adapters** – For displaying dynamic data
- **SharedPreferences** – For session management
- **Content URI Handling** – For image selection
- **Glide Library** – For image loading

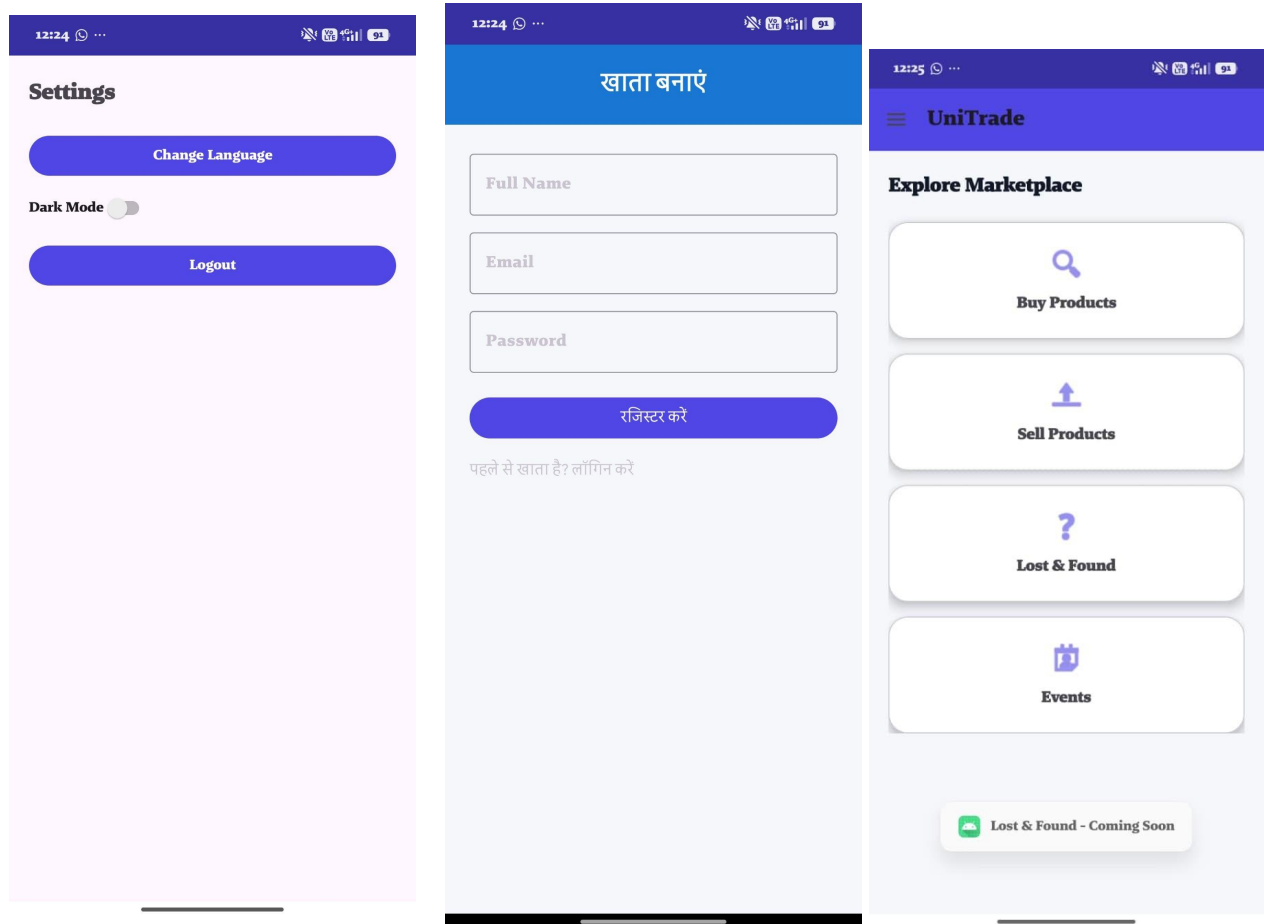
10. Working Flow

1. User registers or logs in
2. Navigates to Home screen
3. Selects Buy or Sell option
4. Views or uploads products
5. Opens product details
6. Clicks “Contact Seller”
7. Chat interface opens
8. Messages are stored and retrieved from database

11. Screen Shots







12. Future Enhancements

- Firebase integration for real-time data
- Online payment gateway
- Push notifications
- Image upload optimization
- Advanced search and recommendation system

13. Conclusion

The UniTrade application successfully demonstrates the implementation of a complete Android-based marketplace system. It integrates authentication, product management, and messaging features using SQLite. The project highlights strong concepts of Android development, UI design, and database handling.